Stroke Syndromes

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Learning Objectives

- Explain the different categories of ischemic and hemorrhagic stroke
- Interpret the symptoms and signs present in stroke patients
- Explain the most frequent syndromes produced by cerebrovascular disease
- Apply the acquired knowledge to clinical examples, with a focus on localization

Contents

What is a Stroke?

- "**Infarct** of the brain, spinal cord, or retinal cell death **attributable to ischemia…**"
- **Broad definition includes:**
  - Ischemic stroke
  - Hemorrhagic stroke
  - 'Silent' strokes
  - Venous infarcts
  - Stroke, NOS

Table 1. Definition of Stroke

The term "stroke" should be broadly used to include all of the following:

Definition of CNS infarction: CNS infarction is brain, spinal cord, or retinal cell death attributable to ischemia, based on

1. pathological, imaging, or other objective evidence of cerebral, spinal cord, or retinal focal ischemic injury in a defined vascular distribution; or
2. clinical evidence of cerebral, spinal cord, or retinal focal ischemic injury based on symptoms persisting ≥24 hours or until death, and other etiologies excluded. (Note: CNS infarction includes hemorrhagic infarctions, types I and II; see "Hemorrhagic Infarction.")

Definition of ischemic stroke: An episode of neurological dysfunction caused by focal cerebral, spinal, or retinal infarction. (Note: Evidence of CNS infarction is defined above.)

Definition of silent CNS infarction: Imaging or neuropathological evidence of CNS infarction, without a history of acute neurological dysfunction attributable to the lesion.

Definition of intracerebral hemorrhage: A focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma.

(Note: Intracerebral hemorrhage includes parenchymal hemorrhages after CNS infarction, types I and II—see "Hemorrhagic Infarction.")

Definition of stroke caused by intracerebral hemorrhage: Rapidly developing clinical signs of neurological dysfunction attributable to a focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma.

Definition of silent cerebral hemorrhage: A focal collection of chronic blood products within the brain parenchyma, subarachnoid space, or ventricular system on neuroimaging or neuropathological examination that is not caused by trauma and without a history of acute neurological dysfunction attributable to the lesion.

Definition of subarachnoid hemorrhage: Bleeding into the subarachnoid space (the space between the arachnoid membrane and the pia mater of the brain or spinal cord).

Definition of stroke caused by subarachnoid hemorrhage: Rapidly developing signs of neurological dysfunction and/or headache because of bleeding into the subarachnoid space (the space between the arachnoid membrane and the pia mater of the brain or spinal cord), which is not caused by trauma.

Definition of stroke caused by cerebral venous thrombosis: Infarction or hemorrhage in the brain, spinal cord, or retina because of thrombosis of a cerebral venous structure. Symptoms or signs caused by reversible edema without infarction or hemorrhage do not qualify as stroke.

Definition of stroke, not otherwise specified: An episode of acute neurological dysfunction presumed to be caused by ischemia or hemorrhage, persisting ≥24 hours or until death, but without sufficient evidence to be classified as one of the above.

CNS indicates central nervous system.

Classification of Strokes

- Ischemic
  - Deprivation of oxygen carrying blood supply rapidly causes damage to neural tissue
  - Arterial occlusion
  - Venous occlusion (see clinical stroke lecture)
  - Hypotension -> hypoperfusion
- Hemorrhagic
  - Intracerebral (intraparenchymal) Hemorrhagic (ICH)
  - Subarachnoid hemorrhage (see clinical stroke lecture)
  - Subdural and epidural hemorrhage (see neurologic emergencies lecture)

Classification of Stroke - Ischemic

- Large vessel occlusion
  - Blockage of a large, named artery such as the middle cerebral artery
  - Can be due to embolism or stenosis leading to occlusion
  - Anterior circulation:
    - ACA and MCA syndromes
  - Posterior circulation:
    - PCA, Basilar artery syndromes

Classification of Stroke - Ischemic

- Small vessel occlusion
  - AKA lacunar infarcts
  - Due to local thrombosis
  - Small penetrating arteries that branch directly from large vessels must handle a large increase in vessel wall stress, making them vulnerable to thrombosis.
  - **Pure motor**, **ataxic hemiparesis**, **pure sensory**, **dysarthria-clumsy hand**, and **sensory-motor** syndromes.

Classification of Stroke - Ischemic

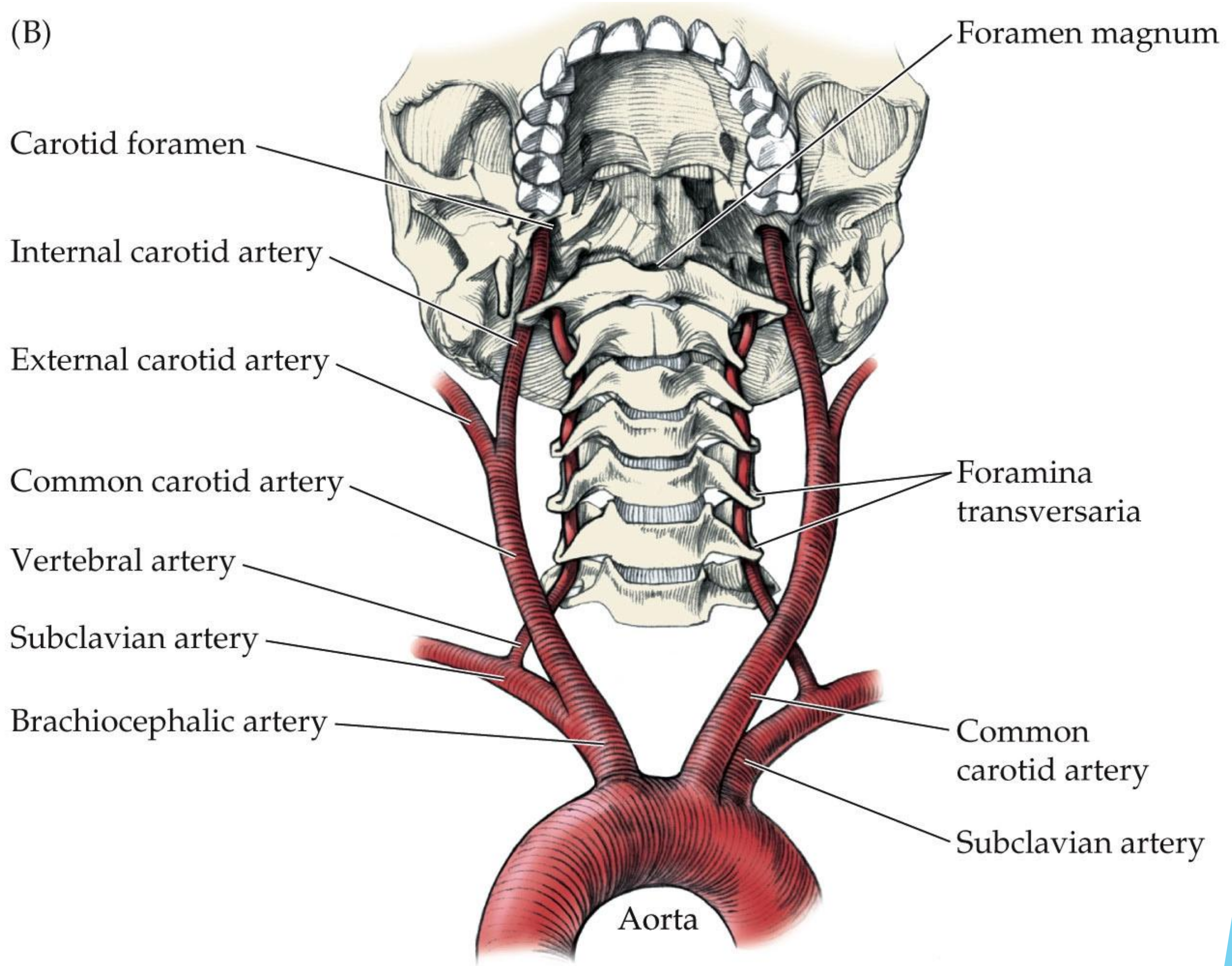
- Hypoperfusion
  - Watershed infarcts
  - Seen in patients with precipitous drops in blood pressure
  - Affects areas of the brain which are in the ‘watershed’ regions between major arterial distributions
  - These areas lack the collateral blood flow that other areas have, making them vulnerable to the effects of low pressure

Classification of Stroke - Hemorrhagic

- Intracerebral Hemorrhage
  - Bleeding withing the parenchyma of the brain
  - Chronic degenerative changes of the penetrating arteries
  - Most common sites are putamen/internal capsule, deep white matter, thalamus, cerebellum, and pons
  - Large blood volume causes mass effect on surrounding tissue, can lead to herniation.

Cerebral Circulation Review

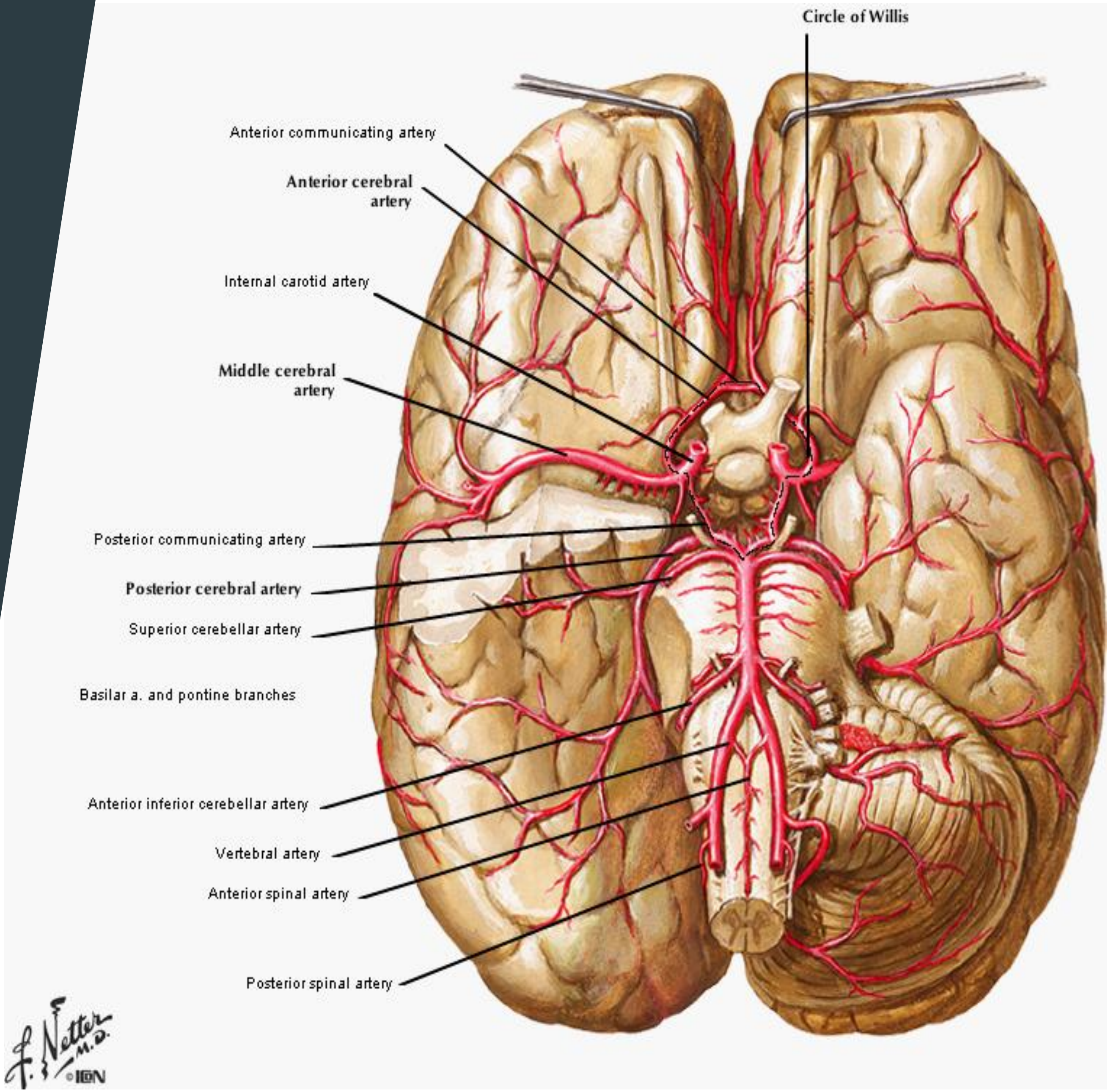
- Anterior circulation begins with the carotid arteries
  - Left common carotid directly off of the aortic arch
  - Right common carotid is a branch of the brachiocephalic trunk
  - The **Internal carotid** supplies the brain
- Posterior circulation begins with the vertebral arteries
  - Bilaterally these are branches off the subclavian



NEUROANATOMY 2e, Figure 10.2 (Part 2)

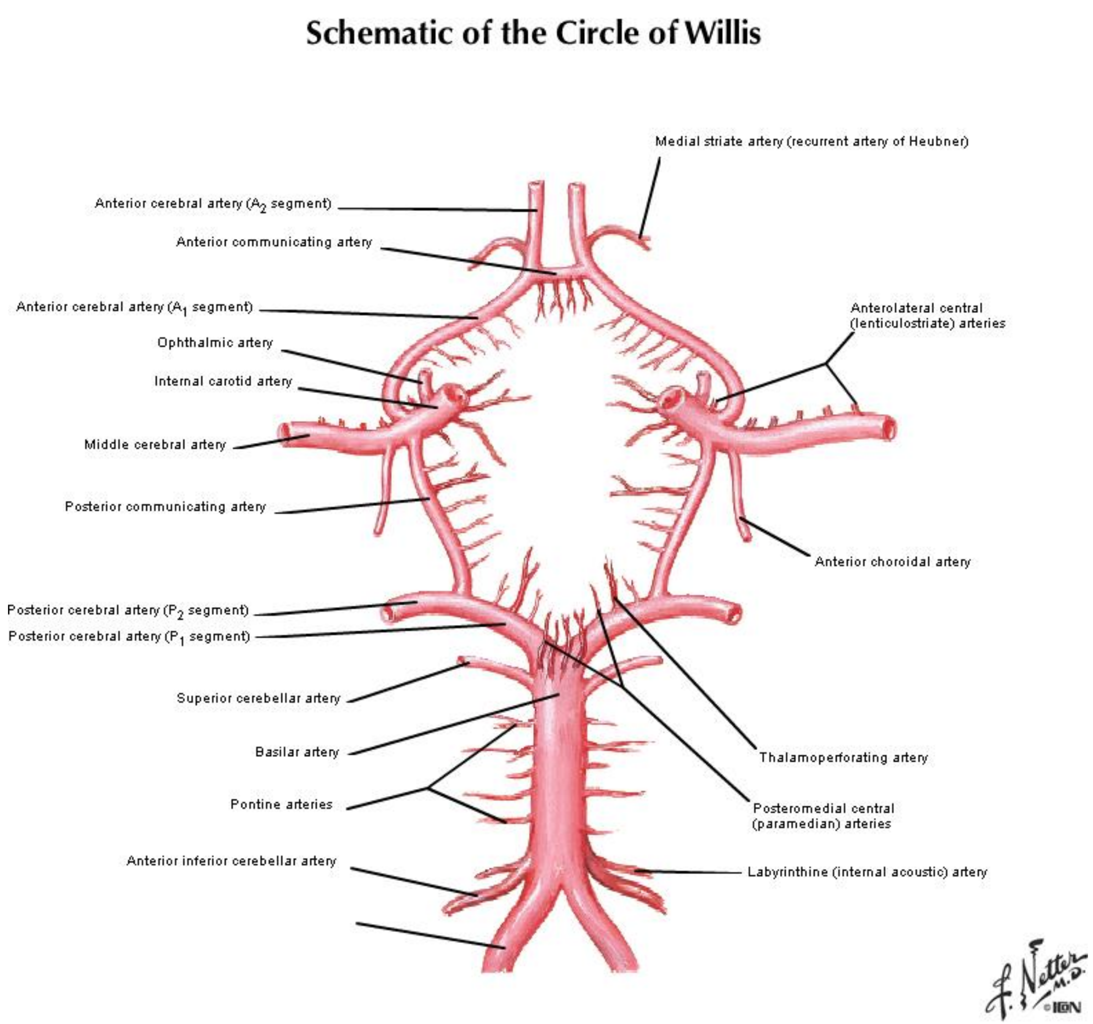
Cerebral Circulation Review

- Anterior circulation – ICA branches:
  - Middle Cerebral Artery
  - Anterior Cerebral Artery
  - Lenticulostriate Arteries
- Posterior circulation:
  - Vertebral Arteries
    - PICA
  - Basilar Artery
    - AICA, SCA
    - Pontine perforating
    - Terminates into bilateral PCAs



Cerebral Circulation Review

- Circle of Willis
  - Provides redundancy to the cerebral circulation
  - Anterior communicating artery connects the two ACAs
  - Posterior communicating artery connects the ICA to ipsilateral PCA
  - HIGHLY VARIABLE
  - Common variant: 'Fetal PCA'
    - No connection between PCA and the basilar, with the PCA instead fed by the PCOM



Cortical Vessel Distributions

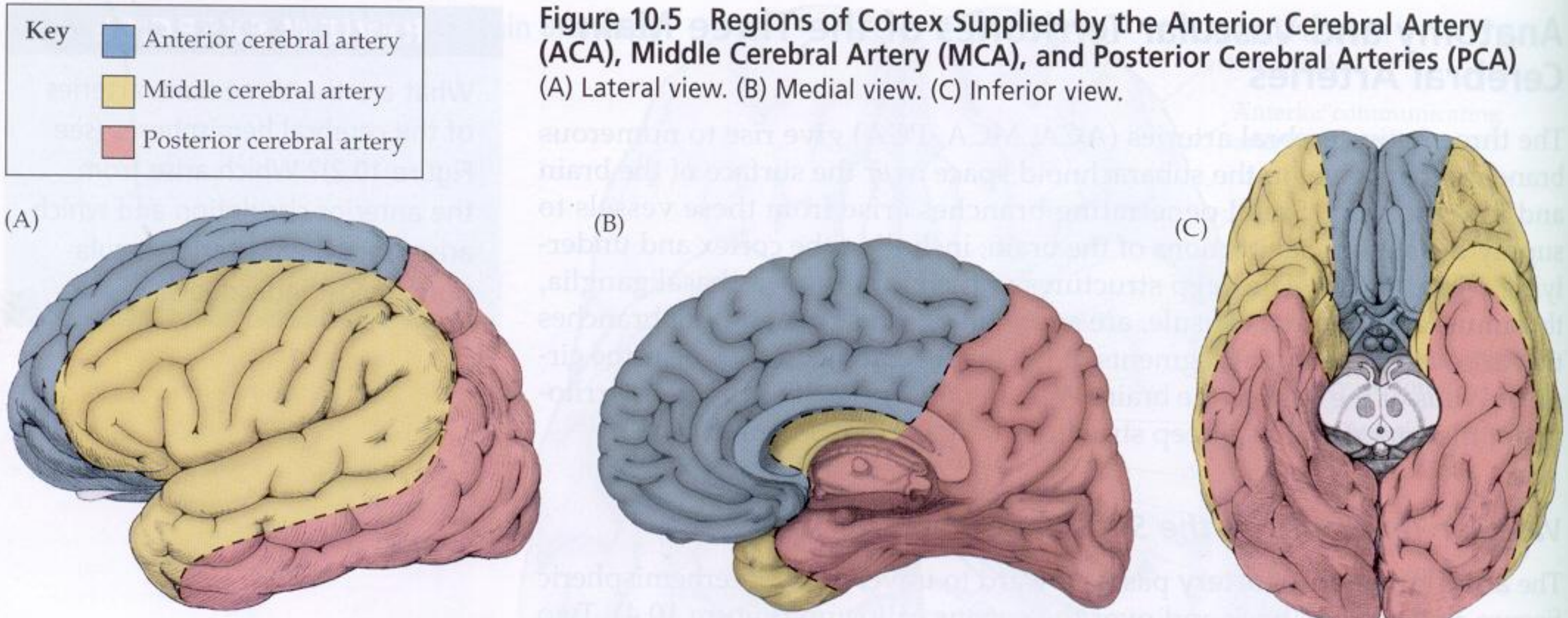


Figure 10.5 Regions of Cortex Supplied by the Anterior Cerebral Artery (ACA), Middle Cerebral Artery (MCA), and Posterior Cerebral Arteries (PCA)
(A) Lateral view. (B) Medial view. (C) Inferior view.

Anterior Large Vessel Syndromes

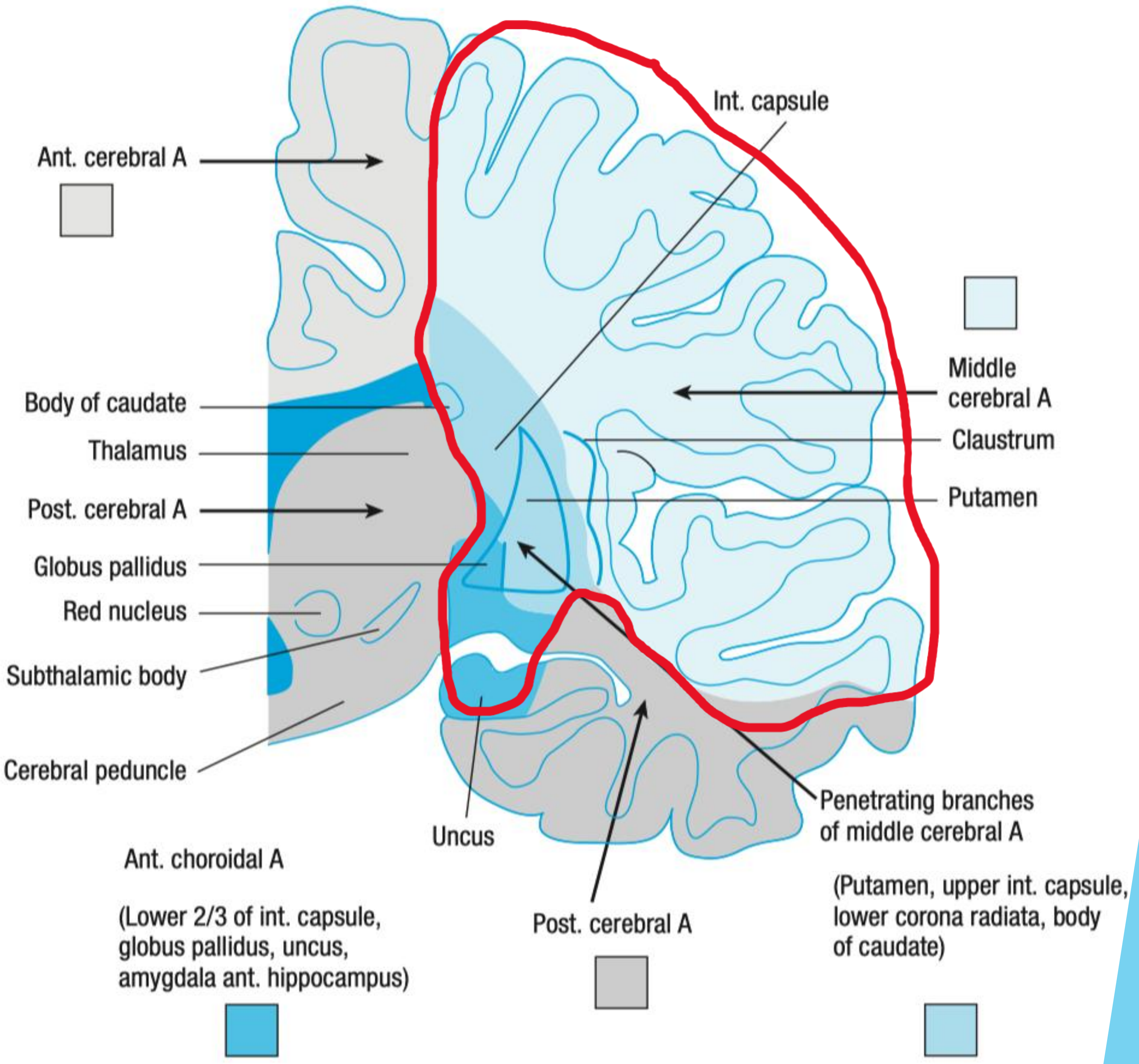
Middle Cerebral Artery Syndromes

- Proximal occlusion
- Distal M1
- Superior M2
- Inferior M2

Anterior Cerebral Artery Syndrome

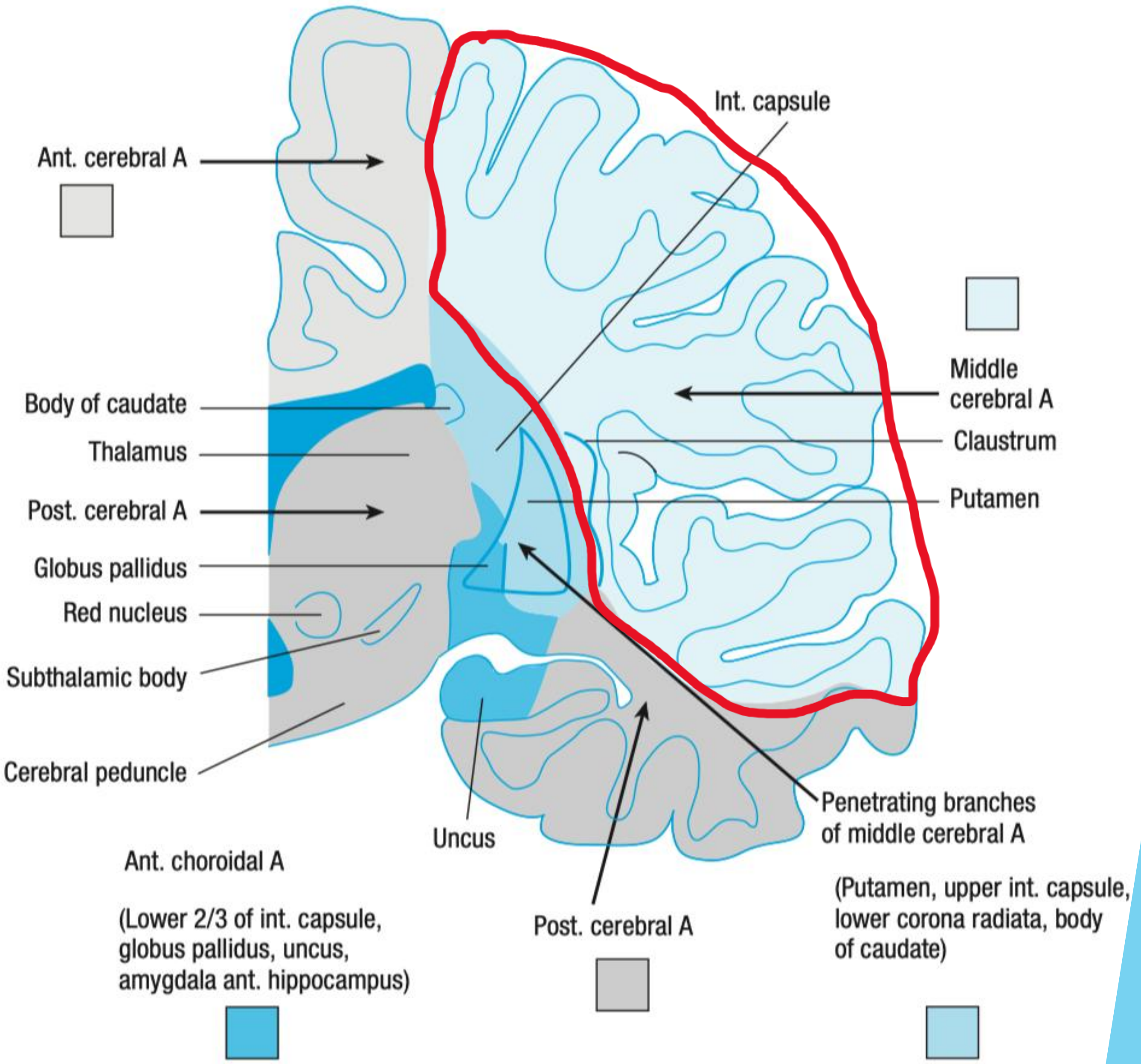
Proximal M1 Occlusion

- Occlusion proximal to the origin of the lenticulostriate arteries
  - Contralateral hemiplegia (body and face)
  - Hemianesthesia
  - Contralateral homonymous hemianopia
  - Ipsilateral gaze preference
  - Global aphasia – if dominant hemisphere
  - Severe hemineglect and anosognosia – if non-dominant hemisphere



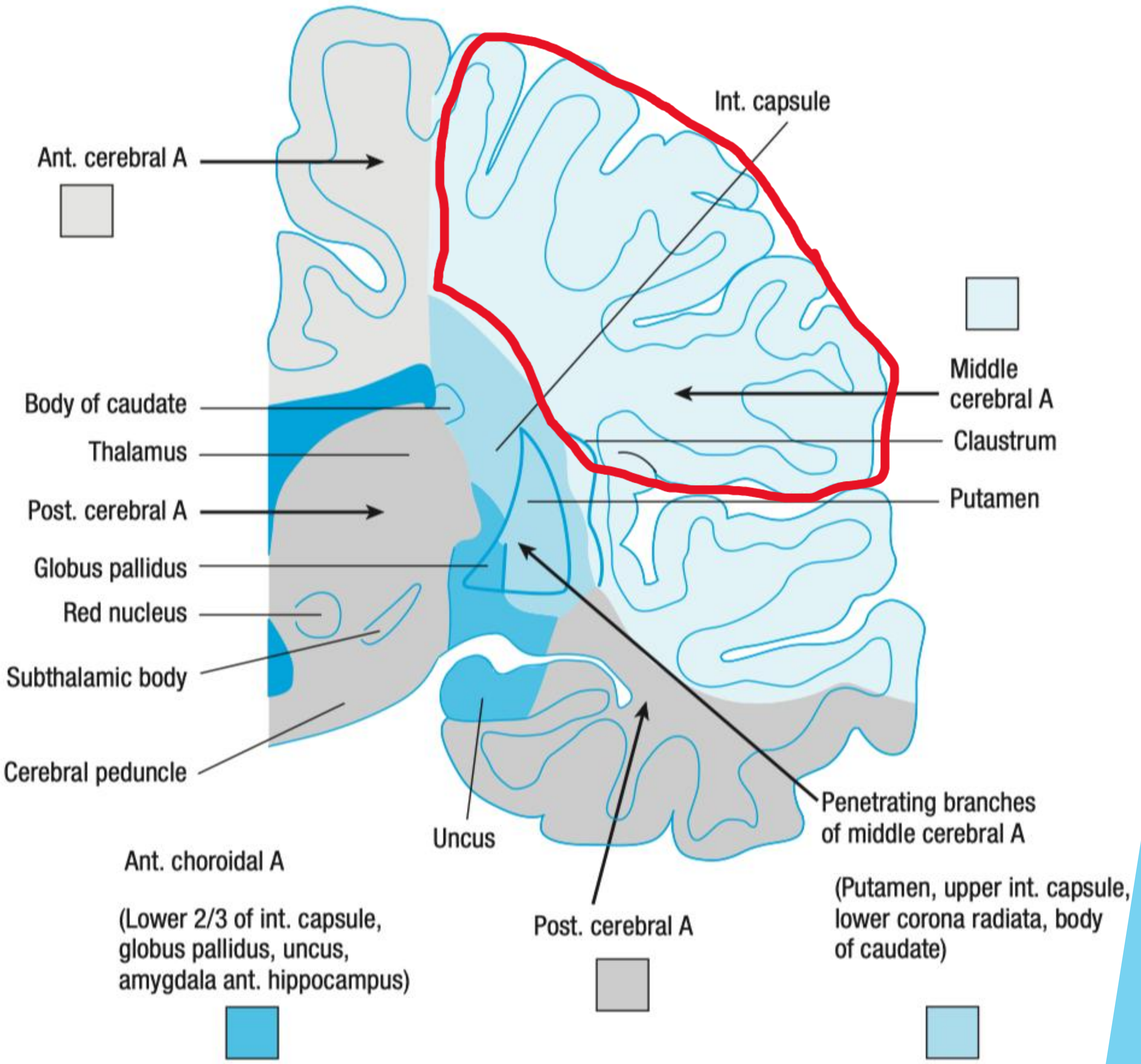
Distal M1 Occlusion

- Occlusion distal to lenticulostriate – spares the internal capsule and other deeper structures
  - Contralateral hemiplegia – **face and arm >> leg**
  - Hemianesthesia
  - Contralateral homonymous hemianopia
  - Ipsilateral gaze preference
  - Global aphasia – if dominant hemisphere
  - Severe hemineglect and anosognosia – if non-dominant hemisphere



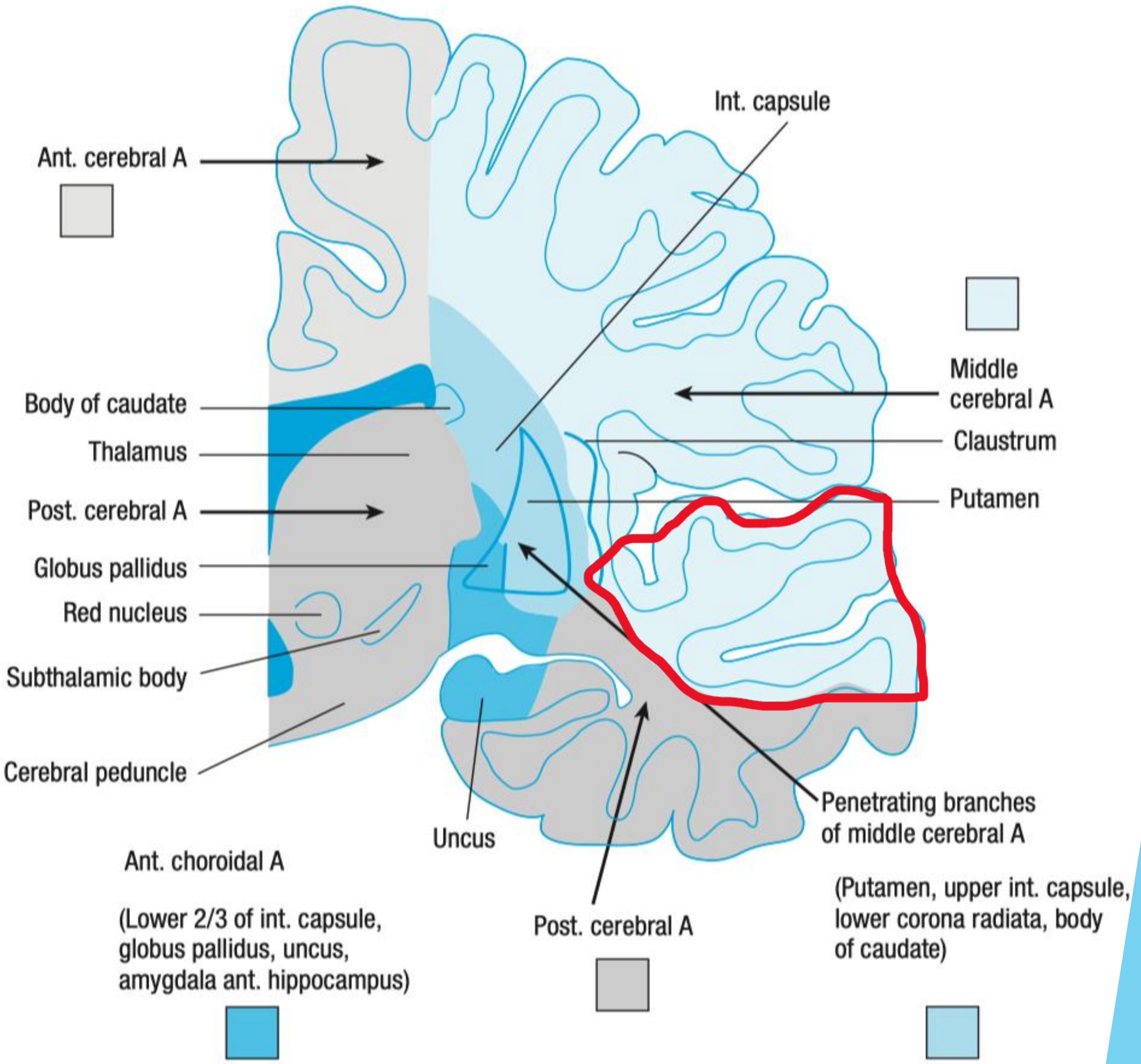
Superior M2

- Beyond first branch of MCA
  - Contralateral hemiplegia – face and arm >> leg
  - Hemianesthesia
  - Contralateral homonymous hemianopia
  - Ipsilateral gaze preference
  - Broca's aphasia – initial global
  - Severe hemineglect and anosognosia – if non-dominant hemisphere



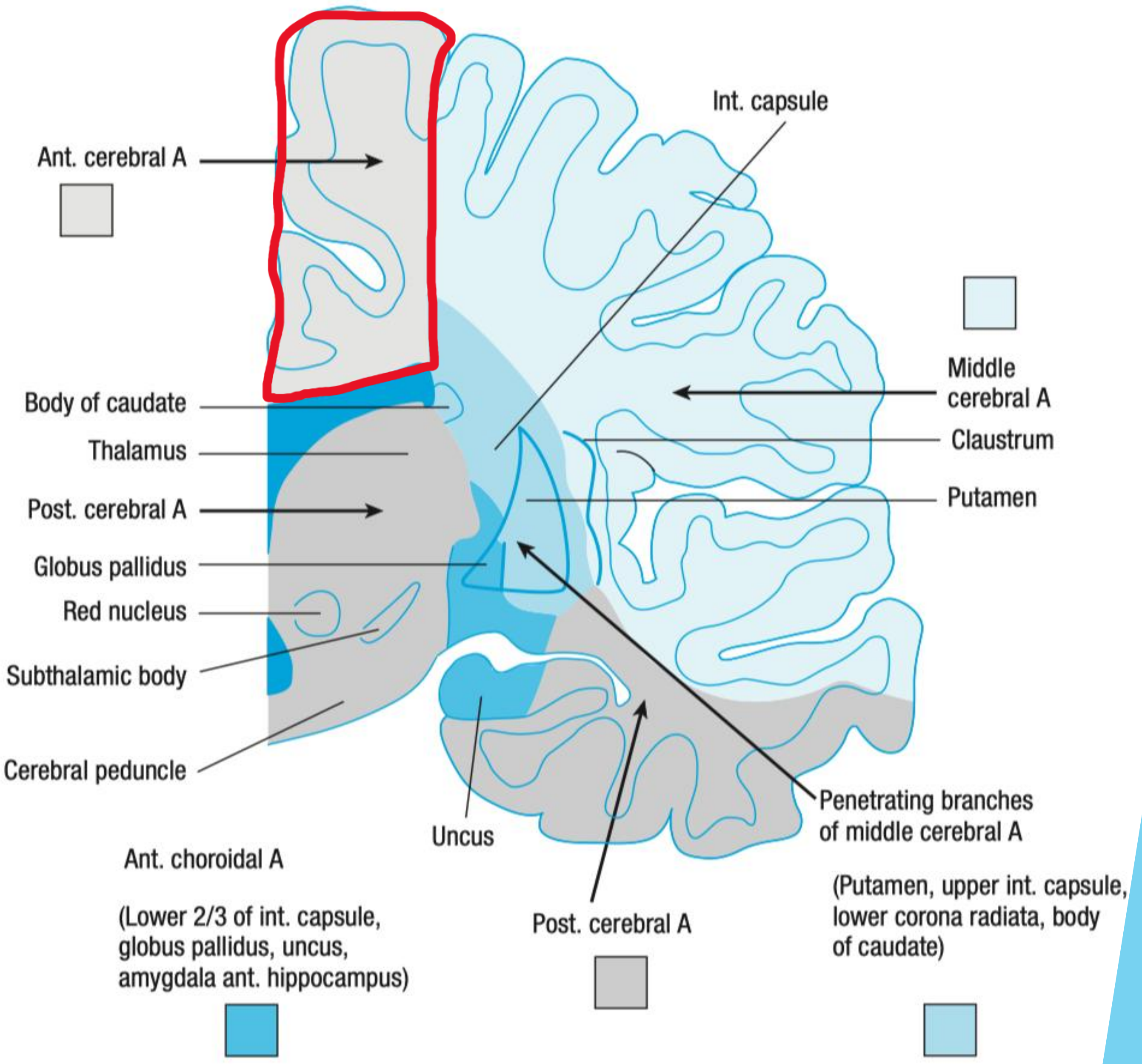
Inferior M2

- Beyond first branch of MCA
  - Less likely to cause contralateral hemiplegia – **face, maybe arm, unlikely to affect lower extremities**
  - Can cause contralateral superior quadrantopia
  - No gaze preference
  - **Wernicke's aphasia** – initial global



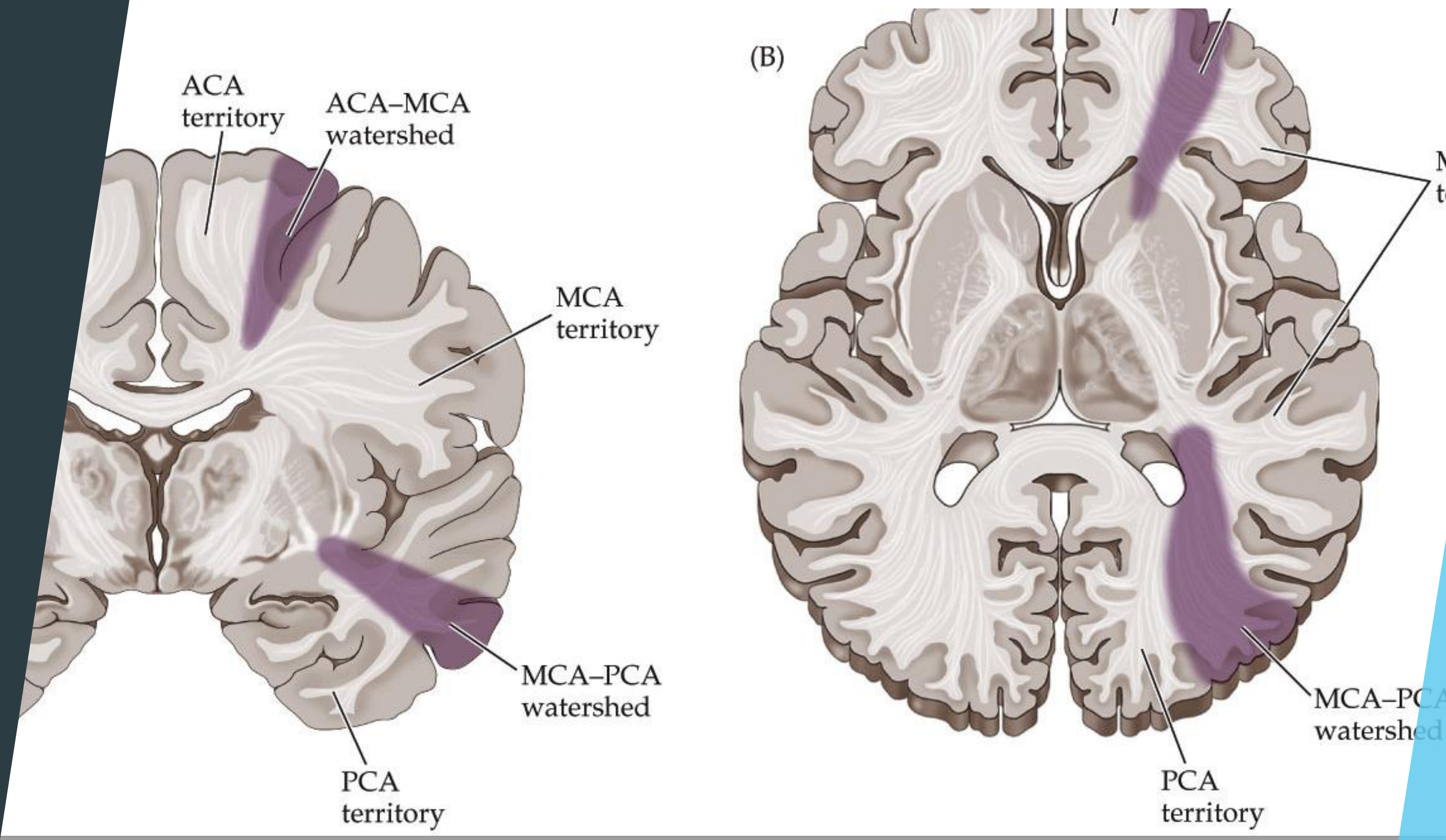
ACA Syndrome

- Infarcts are usually only seen if occlusion occurs *distal* to anterior communicating artery
- Contralateral sensorimotor deficit in foot and leg
- Can cause a transcortical motor aphasia
- Urinary incontinence
- Contralateral paratonia
- Contralateral apraxia



Watershed Infarcts

- Vascular territories overlap zones
- ACA-MCA
  - Associated with transcortical motor aphaisa
- MCA-PCA
  - Associated with transcortical sensory aphasia
- Often seen after hypotensive events or in patients with many vascular risk factors after surgery

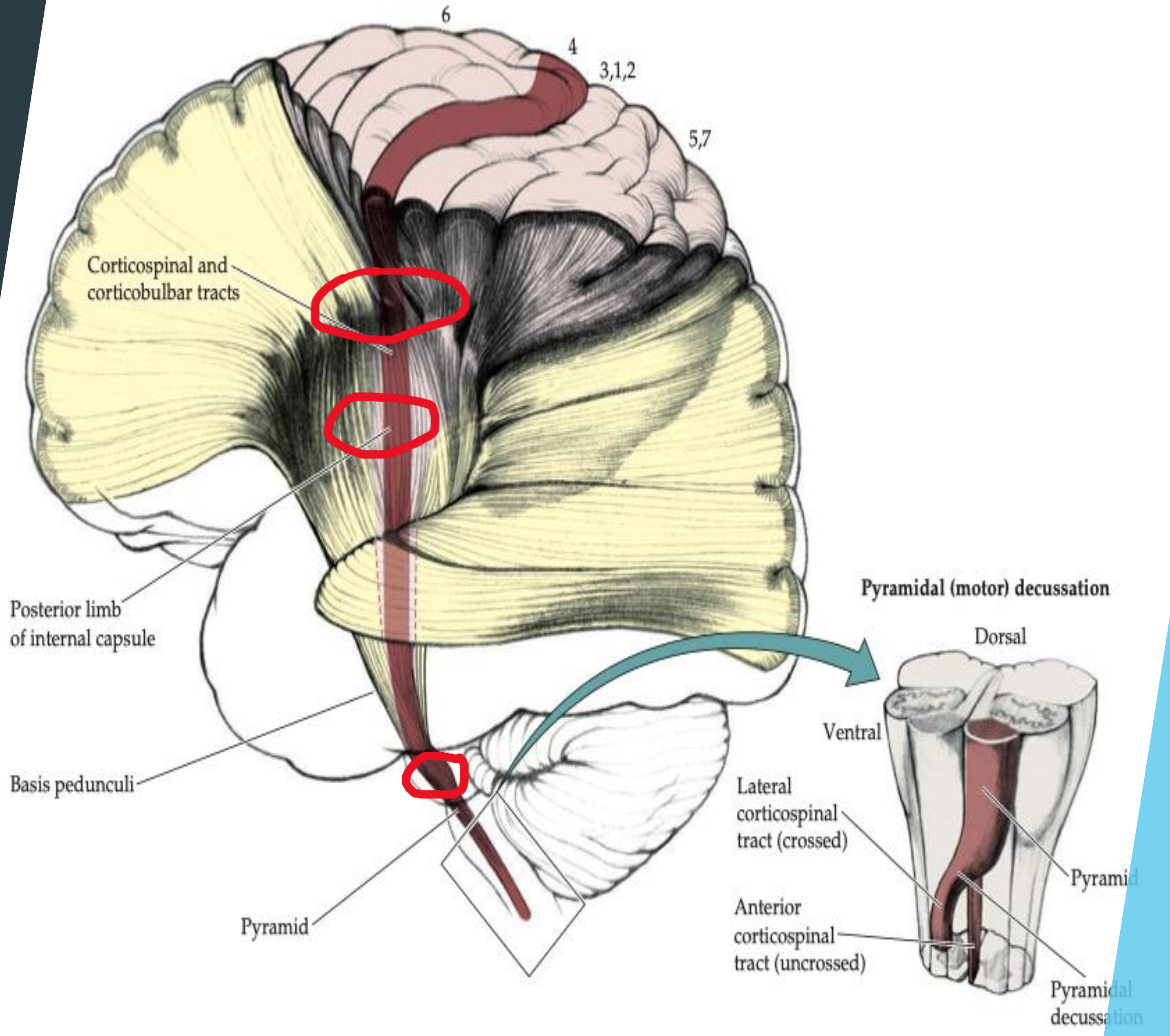


Small Vessel Syndromes

- Pure Motor Stroke
- Pure Sensory Stroke
- Mixed Sensorimotor Stroke
- Dysarthria-clumsy Hand Syndrome
- Ataxic Hemiparesis

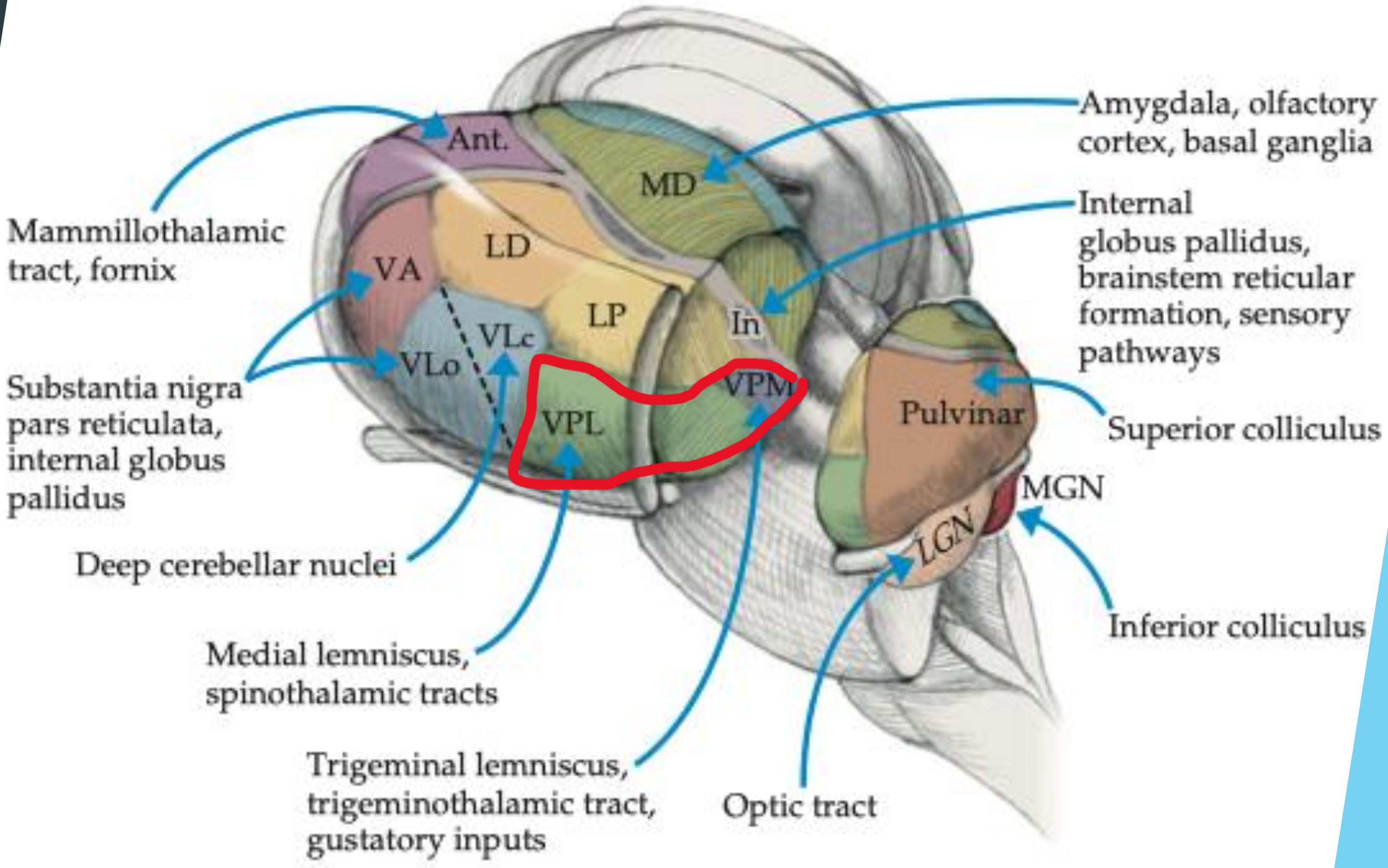
Pure Motor Stroke

- Contralateral Hemiparesis
  - Equal in arms/legs/face
- No sensory deficits
- No cortical deficits
  - No neglect
  - No aphasia
  - No apraxia
- Localization:
  - Basis Pontis
  - Posterior Limb Internal Capsule
  - Corona Radiata



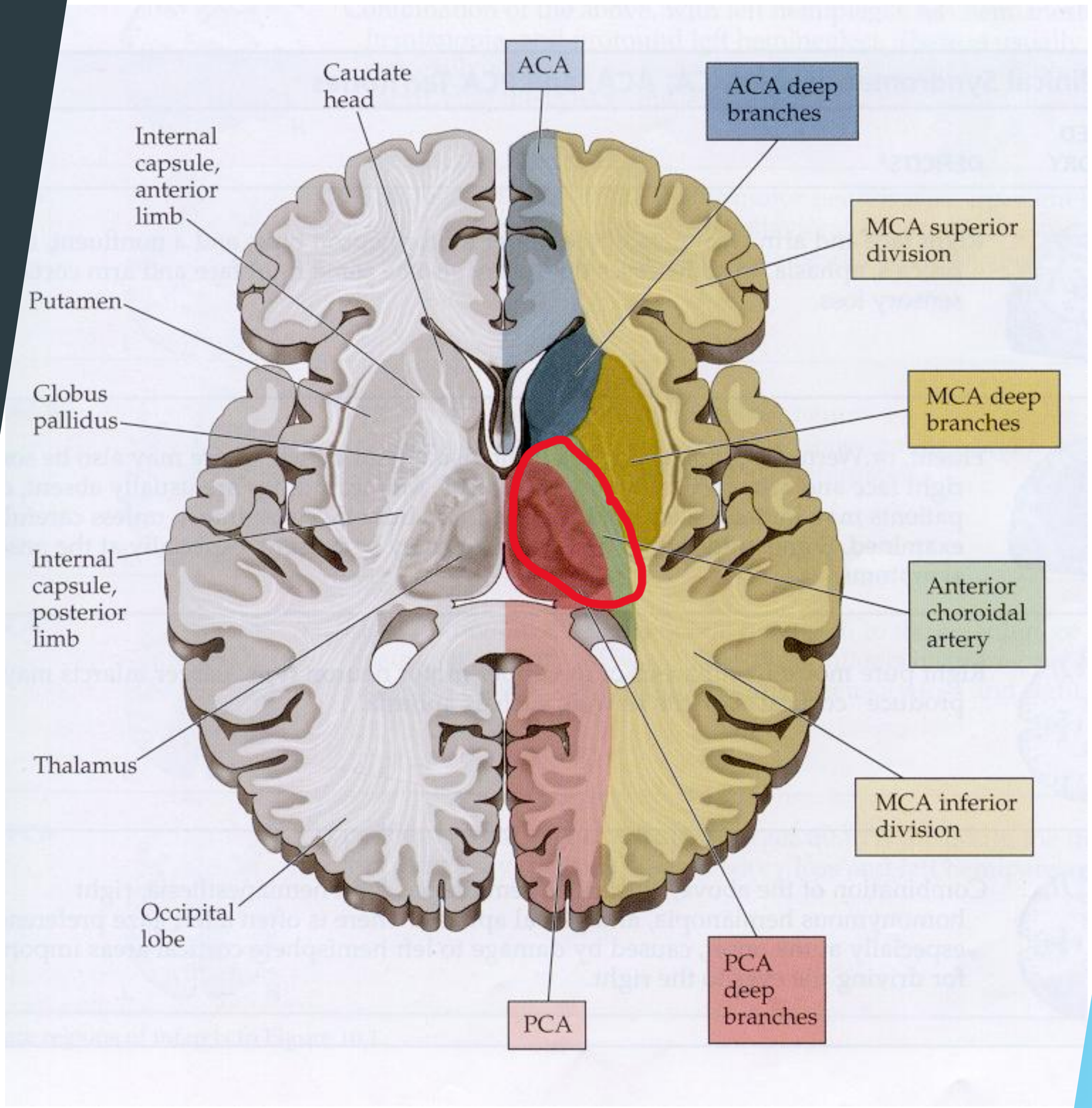
Pure Sensory Stroke

- Contralateral Hemianesthesia
  - Equal in arms/legs/face
- No motor deficits
- No cortical deficits
- Associated with thalamic pain syndrome
  - Onset of hemibody pain days to weeks after stroke
- Localization:
  - VPL, VPM of thalamus



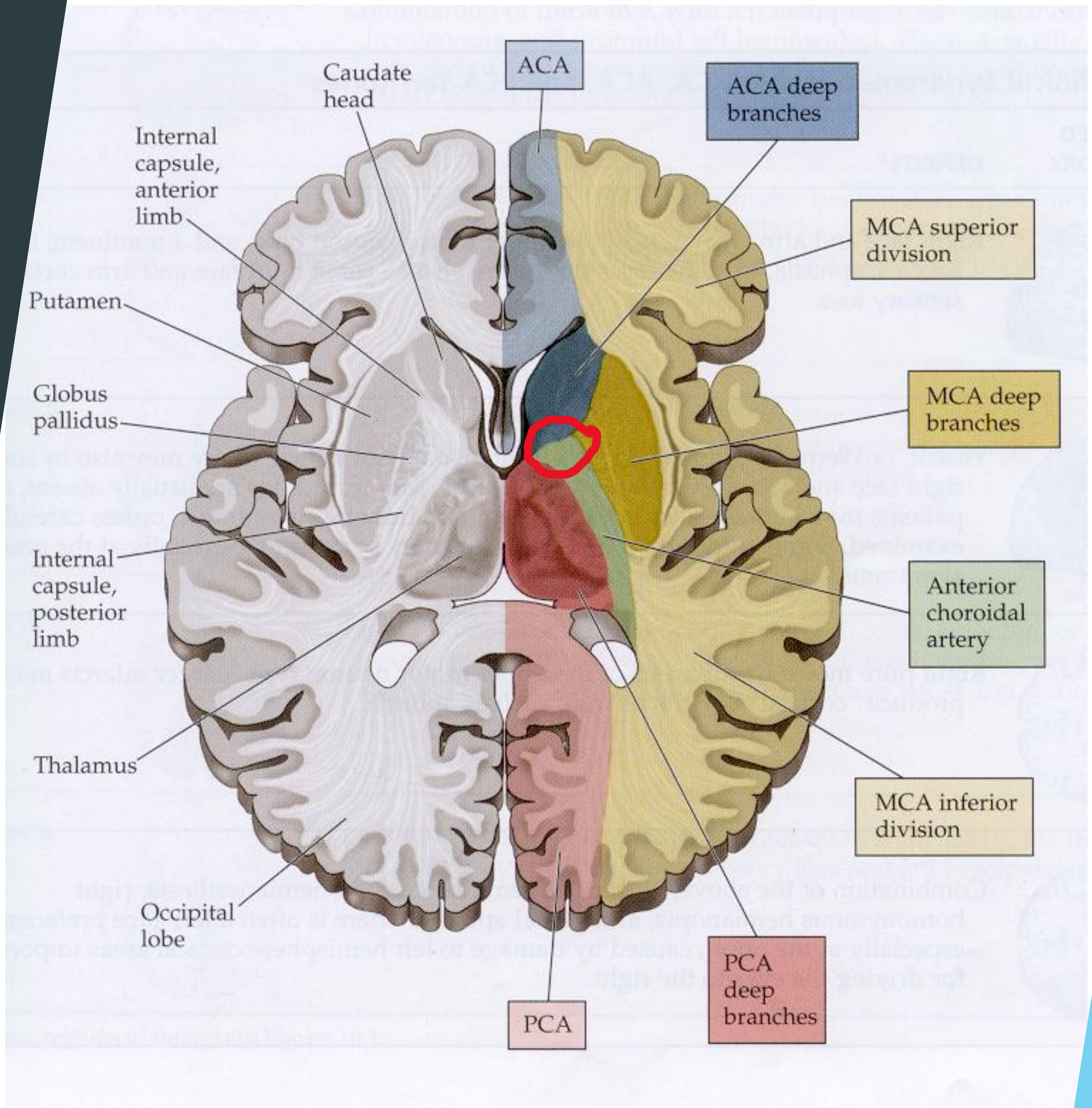
Mixed Sensorimotor

- **Contralateral Hemiparesis and Hemisensory loss**
  - Equal in arms/legs/face
- No cortical deficits
- Localization:
  - Thalamus plus internal capsule



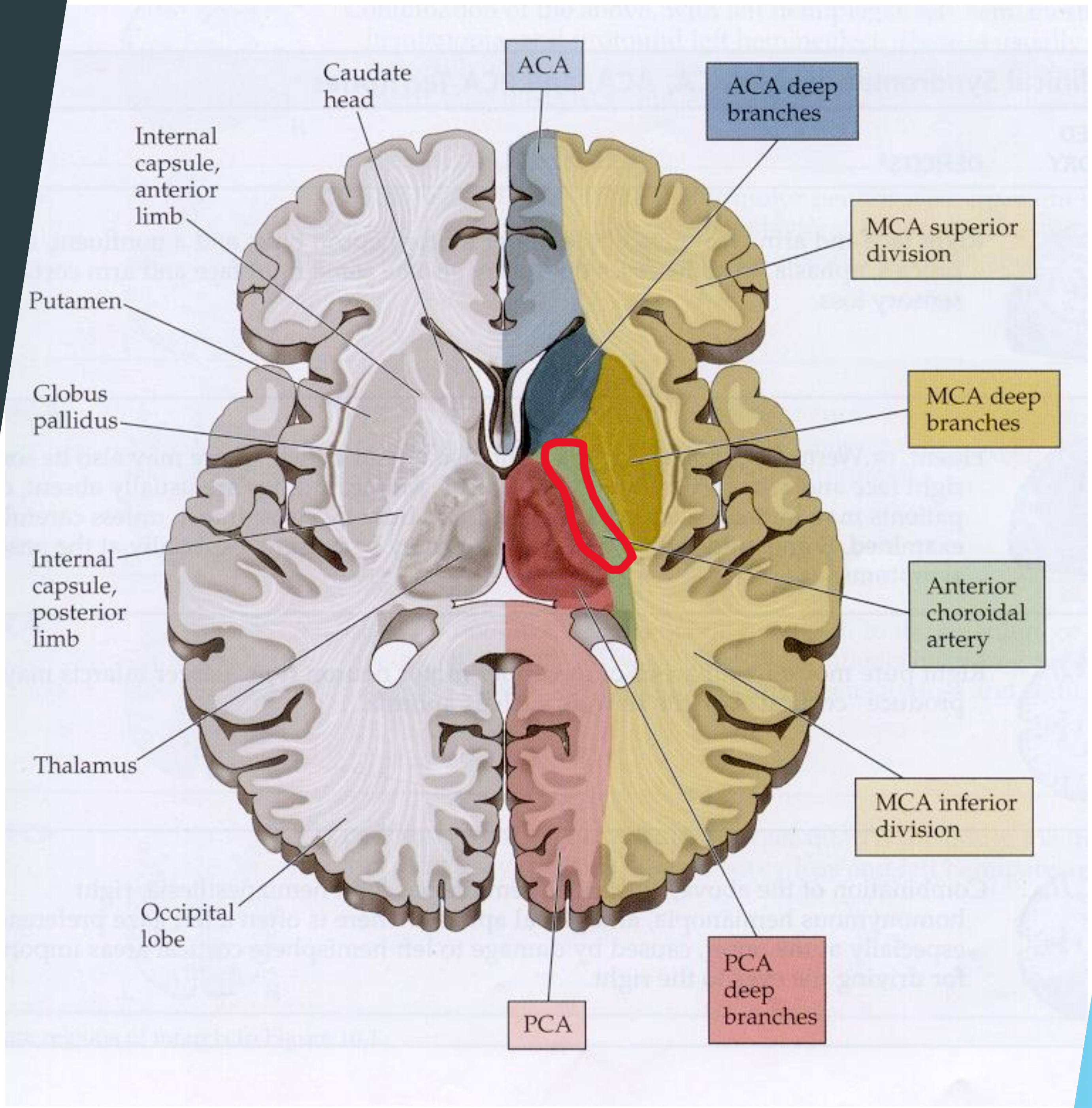
Dysarthria-clumsy Hand

- Dysarthria
- Contralateral Hand Weakness
- No cortical deficits
- Localization:
  - Genu of Internal Capsule
  - Basis Pontis



Ataxic Hemiparesis

- Contralateral Hemiparesis
- Contralateral Ataxia
- No cortical deficits
- Localization:
  - Internal Capsule
  - Basis Pontis

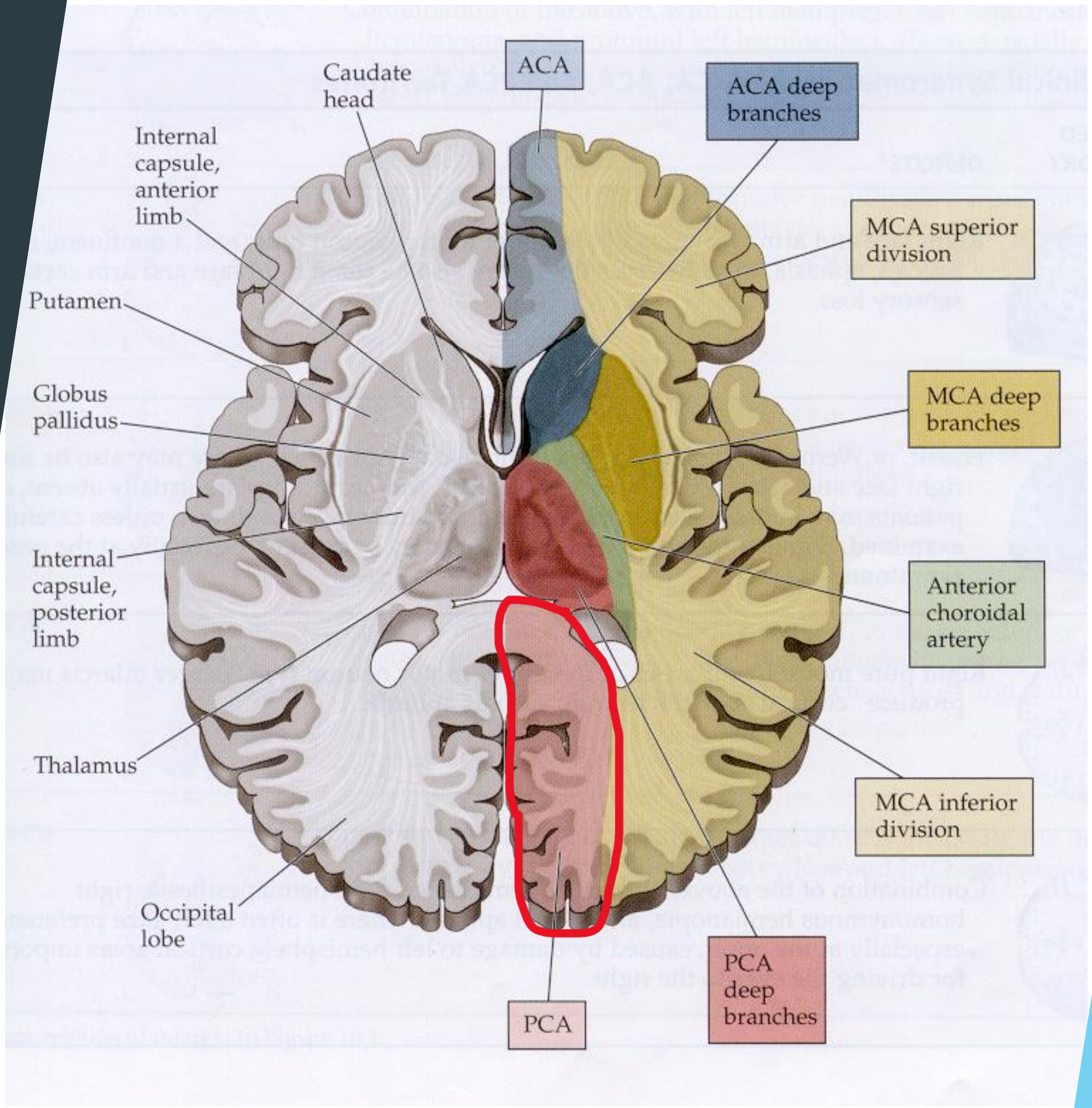


Posterior Circulation Syndromes

- PCA syndrome
- Basilar Occlusion
- Brainstem stroke syndromes
  - Rule of 4s

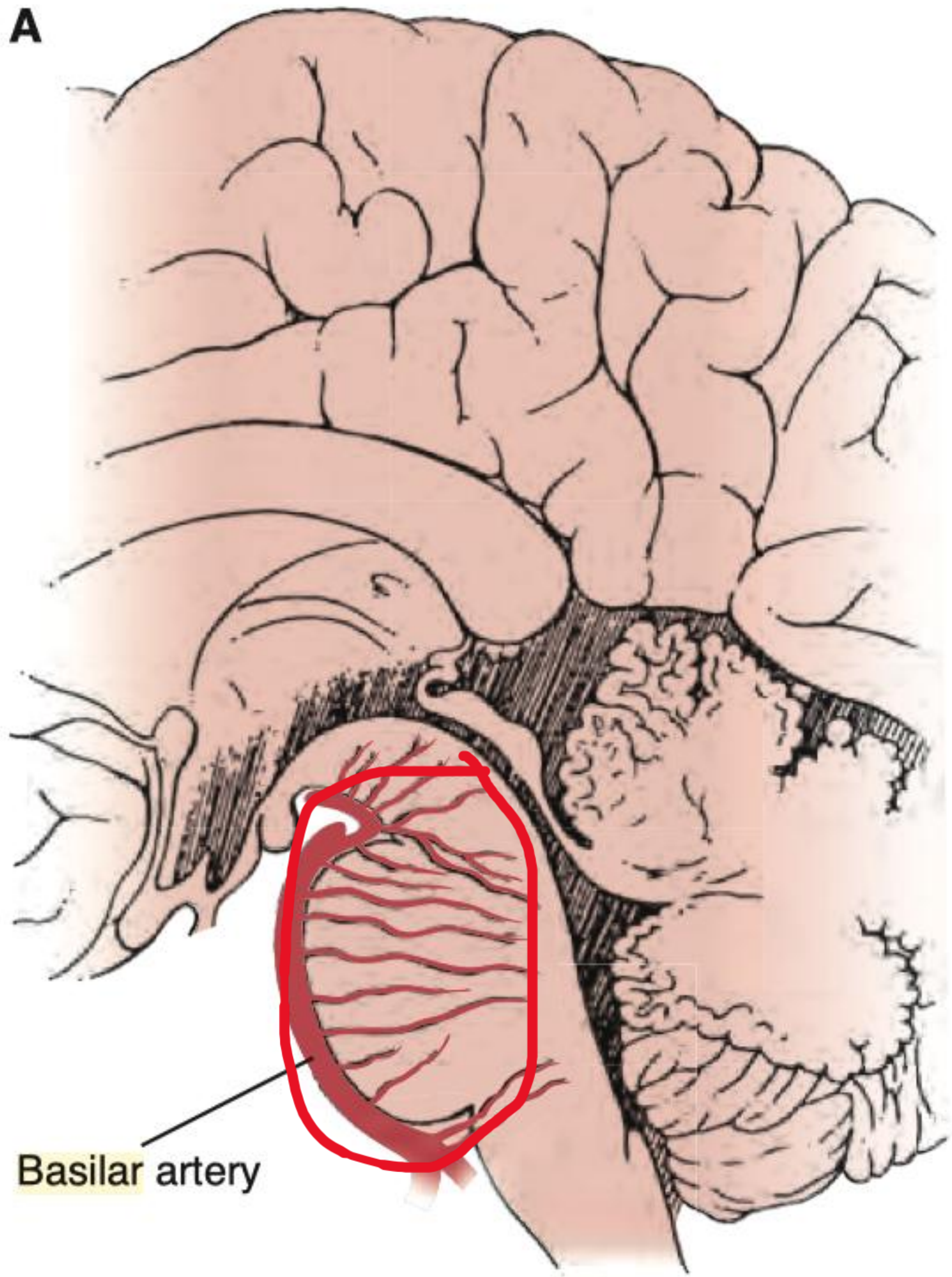
PCA Stroke

- Contralateral Hemianopia
- If bilateral, cortical blindness can occur
  - More rarely Anton's syndrome – confabulation of sight
- Proximal PCA occlusion can affect peduncular perforating branches of PCA:
  - Midbrain structures
  - 'Weber syndrome' – contralateral hemiplegia + ipsilateral oculomotor palsy



Basilar Thrombosis

- Devastating injury to midbrain and pontine structures
- Associated with **‘Locked in syndrome’**
  - Patient retains awareness but complete paralysis, with intact sensation
  - In some cases, only able to make vertical eye movements



Brainstem Stroke Syndromes

- The named brainstem stroke syndromes can be overwhelming
- The boards (and my exam questions) will always give more than the eponym, either location or blood vessel as an answer choice
- A simplified roadmap for dealing with brainstem vascular localization has been proposed by Peter Gates, and Australian neurologist. This is widely used in boards prep
- This system utilizes a focus on certain deficits that combine to make a reliable localization

Brainstem Stroke Syndromes

Rule of 4

Why it works

- The blood supply of the brainstem can be divided into:
  - Paramedian perforating arteries from the vertebral and basilar artery supplying medial aspects of the brainstem
  - Circumferential branches that reach the lateral aspects of the brainstem
- Four cranial nerves arise from the medulla (12, 11, 10, 9,) and four from the pons (8, 7, 6, and 5,) plus two from the midbrain (4 and 3.)
- By utilizing your knowledge of select structures within the brainstem you will be able to localize a brainstem lesion to the level (medulla, pons, midbrain,) and whether it is a lateral syndrome or medial syndrome.

Brainstem Stroke Syndromes

Rule of 4

How it works

- Paramedian Brainstem contains sets of structures, which can be remembered as starting with ‘M’:
  - Motor pathways (CST)
  - Medial Lemnisci (vibration, proprioception)
  - Medial Longitudinal Fasciculus
  - Motor nuclei that divide evenly into 12 (3, 4, 6, 12.) All the rest are lateral brainstem
- Lateral Brainstem contains structures which can be remembered as starting with ‘S’:
  - Spinothalamic pathway
  - Sympathetic pathway
  - Sensory nucleus of CNV
  - Spinocerebellar tract
  - Plus all the motor nuclei that don’t evenly divide into 12

Brainstem Stroke Syndromes

Rule of 4

How it works

- Step 1: Recognize the long tract findings to determine if the lesion is medial or lateral. For example:
  - Lateral lesions cause crossed sensory findings due to the combination of spinothalamic and CNV sensory nucleus. Pain and temp would be contralateral in the body and ipsilateral in the face.
  - Corticospinal tract findings suggest a paramedian localization.
  - Ataxia (spinocerebellar tract) that is opposite pain and temperature loss from the body localizes laterally.
- Step 2: Determine level of brainstem by considering cranial nerve involvement. For example:
  - Oculomotor palsy with CST findings suggests medial midbrain
  - Adding a facial nerve palsy to the third example above would give a lateral pontine localization

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Right sided UMN weakness of the face, arm, and leg
  - Left sided oculomotor palsy
- Medial or Lateral?
- Midbrain, Pons, or Medulla?

Brainstem Stroke Syndromes

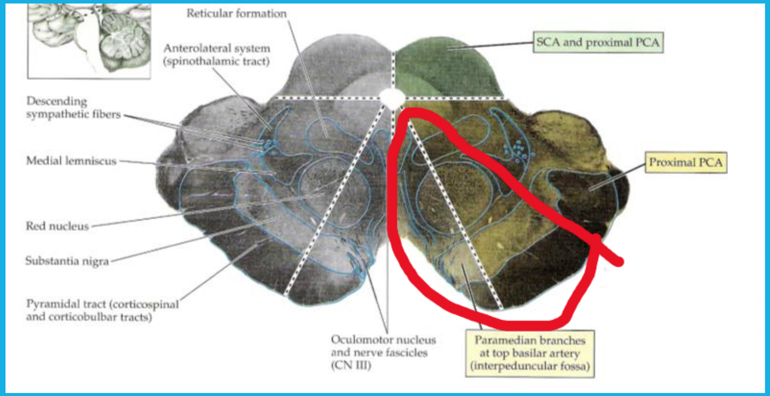
Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Right sided UMN weakness of the face, arm, and leg
  - Left sided oculomotor palsy
- Medial or Lateral?
- Midbrain, Pons, or Medulla?
- Left medial midbrain

Brainstem Stroke Syndromes

Weber Syndrome -Medial Midbrain



- Contralateral Face/Arm/Leg weakness
- Ipsilateral Oculomotor Palsy
- Usually due to peduncular perforating branches off PCA

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Slurred speech
  - Right sided LMN pattern facial weakness
  - Right sided arm and leg ataxia
  - Right sided Horner's
  - Left sided pain and temp loss for body
  - Right sided pain and temp loss for face
- Medial or Lateral?
- Midbrain, Pons, or Medulla?

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Slurred speech
  - Right sided LMN pattern facial weakness
  - Right sided **arm and leg** ataxia
  - Right sided Horner's
  - Left sided pain and temp loss for body
  - Right sided pain and temp loss for face
- Medial or Lateral?
- Midbrain, Pons, or Medulla?
- Right lateral pontine

Brainstem Stroke Syndromes

Lateral Pontine Syndrome - Marie-Foix syndrome

- Deficits are variable due to the AICA and PICA crossover
- Usually, an AICA syndrome
- Additional and variable deficits can include:
  - Nausea and vomiting
  - Vertigo – horizontal and vertical nystagmus
  - Tinnitus
  - Deafness
- AKA Marie-Foix syndrome

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Rightward gaze palsy
  - Left sided UMN body weakness
  - Left sided body vibration/proprioception loss
- Medial or Lateral?
- Midbrain, Pons, or Medulla?

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Rightward gaze palsy
  - Left sided UMN body weakness
  - Left sided body vibration/proprioception loss

- Medial or Lateral?
- Midbrain, Pons, or Medulla?
- Medial Pontine syndrome

Brainstem Stroke Syndromes

Medial Pontine Syndrome

- Due to occlusion of the paramedian pontine perforating arteries
- Deficits include:
  - Contralateral hemiplegia of body
  - Ipsilateral horizontal gaze palsy (CN6)
  - Contralateral vibration/proprioception loss (Medial lemniscus)
  - Can cause an INO due to involvement of the medial longitudinal fasciculus

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Left sided UMN body weakness
  - Left sided body vibration/proprioception loss
  - Right sided tongue weakness (tongue deviates towards the right)
- Medial or Lateral?
- Midbrain, Pons, or Medulla?

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Left sided UMN body weakness
  - Left sided body vibration/proprioception loss
  - Right sided tongue weakness (tongue deviates towards the right)
- Medial or Lateral?
- Midbrain, Pons, or Medulla?
- Medial Medullary Syndrome

Brainstem Stroke Syndromes

Medial Medullary Syndrome – Alternating Hemiplegia

- Due to occlusion of the paramedian branches of the vertebrals or the anterior spinal artery
- Called alternating hemiplegia because of the crossed motor deficits
- Deficits include:
  - Contralateral hemiplegia of body
  - Contralateral vibration/proprioception loss (Medial lemniscus)
  - Ipsilateral LMN tongue weakness (CN12)

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Right sided ataxia
  - Left sided body pain and temperature sensation loss
  - Right sided facial pain and temperature sensation loss
  - Right sided Horner's syndrome
  - Dysarthria
  - Flattened palate on the right
- Medial or Lateral?
- Midbrain, Pons, or Medulla?

Brainstem Stroke Syndromes

Rule of 4

Practice

- A 65 year old man with all the vascular risk factors presents with sudden onset:
  - Right sided ataxia
  - Left sided body pain and temperature sensation loss
  - Right sided facial pain and temperature sensation loss
  - Right sided **Horner's** syndrome
  - Dysarthria
  - Flattened palate on the right
- Medial or Lateral?
- Midbrain, Pons, or Medulla?
- Right Lateral Medulla

Brainstem Stroke Syndromes

Lateral Medullary Syndrome – Wallenberg's

- Due to occlusion of the vertebral or PICA
- Deficits include:
  - No CST findings
  - Ipsilateral nucleus ambiguous lesion - flattened palate, dysarthria
  - Ipsilateral ataxia
  - Ipsilateral pain and temp loss in the face
  - Contralateral pain and temp loss in the body